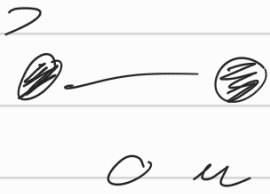


# 7. drá

## BFS



$$d(s) := 0$$

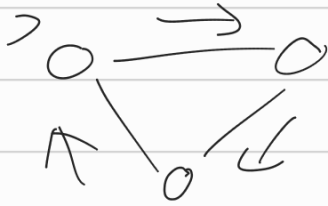
$$\pi(s) := \emptyset$$

-  $d$ : út hozza

$$d(u) := \infty$$

-  $\pi$ : vissza hív

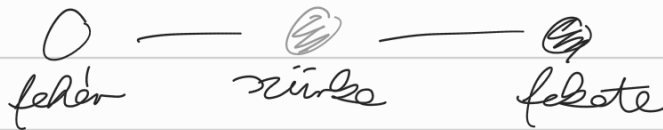
$$\pi(u) := \emptyset$$



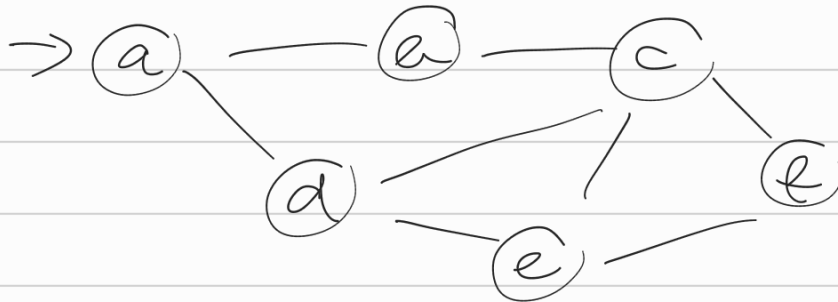
- fehér: felfedezett

- sárga: elért, de nem feldolgozott

- fekete: feldolgozott

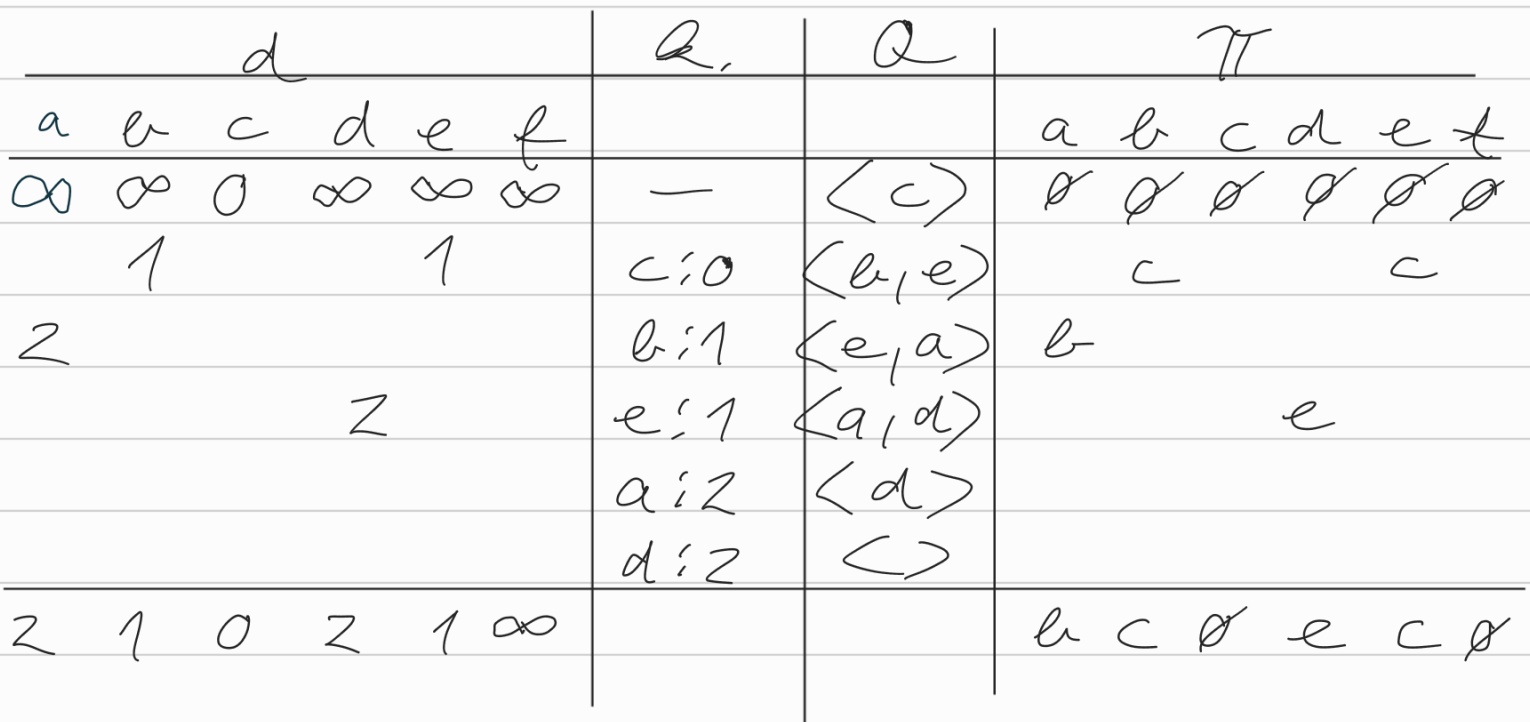


## Feladat:



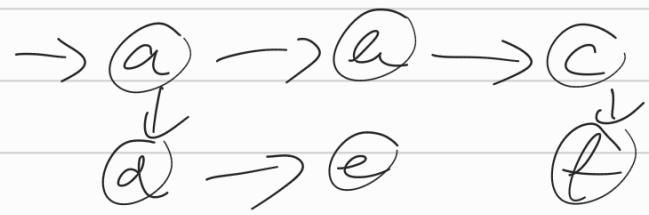
| d |   |   |   |   |   | Elérhető |       | π |   |   |   |   |   |
|---|---|---|---|---|---|----------|-------|---|---|---|---|---|---|
| a | b | c | d | e | f | csúcs    | Queue | a | b | c | d | e | f |
| 0 | ∞ | ∞ | ∞ | ∞ | ∞ | —————    | <a>   | ∅ | ∅ | ∅ | ∅ | ∅ | ∅ |
| 1 |   |   | 1 |   |   | a:0      | <b,d> | a |   |   | a |   |   |
|   | 2 |   |   |   |   | b:1      | <d,c> |   | b |   |   |   |   |
|   |   |   |   | 2 |   | d:1      | <c,e> |   |   |   |   | d |   |
|   |   |   |   |   | 3 | c:2      | <e,f> |   |   |   |   |   | c |

0 1 2 1 2 3



|     |   |   |   |   |   |   |
|-----|---|---|---|---|---|---|
| ms  | a | b | c | d | e | f |
| ms' | Ø | a | b | a | d | c |

- f - be az út:  $\langle a, b, c, f \rangle$

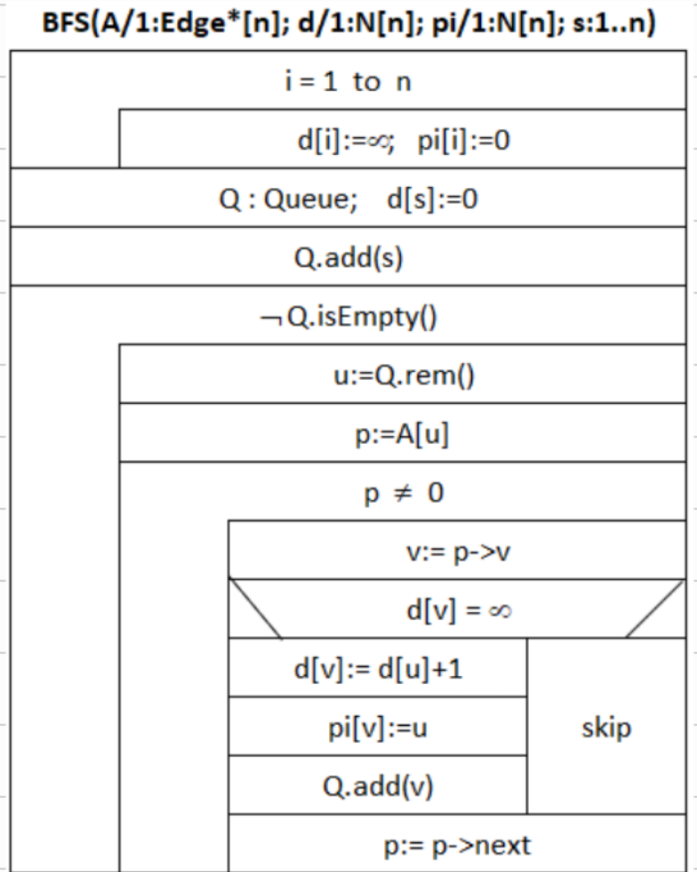
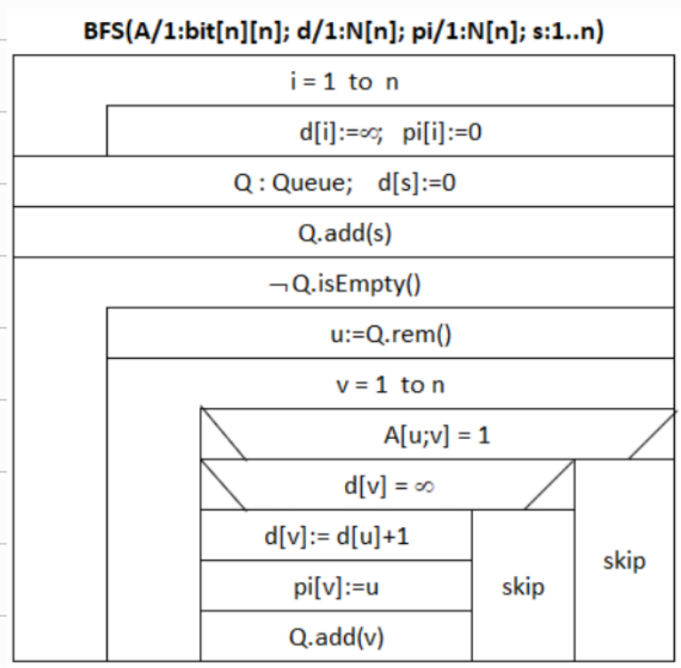


Feladat:

| Csúcs | 1  | 2 | 3 | 4 | 5  | 6  | 7 | 8 | 9 | 10 | 11 | 12 |
|-------|----|---|---|---|----|----|---|---|---|----|----|----|
| súly  | 10 | 4 | 6 | 0 | 11 | 10 | 2 | 7 | 6 | 4  | 7  | 10 |

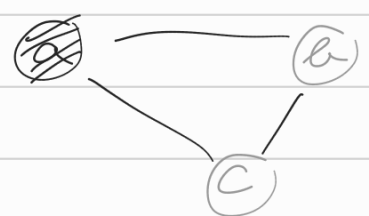
- 5 - ösbe az út:  $\langle 4, 2, 7, 11, 5 \rangle$

BFS ellistával és csúcmátrixal:



Gnát fa - e i

- fa, ha összefüggő és biconnected



- ha egy csúcs súlyos, akkor nem lehet súlyos nanscedja

- összefüggő, ha a feldolgozott csúcsok száma megegyezik a gráf csúcsainak a számával

| Fa_e(A/1:Edge*[n]) : Bool                              |                                             |                |
|--------------------------------------------------------|---------------------------------------------|----------------|
| color:szin[1..n] //color, szin típusú tömb létrehozása |                                             |                |
| i = <del>2</del> <sup>1</sup> to n                     |                                             |                |
| color[i]:= white                                       |                                             |                |
| Q : Queue color[1]:=grey                               |                                             |                |
| Q.add(1) <del>fa: igaz</del> c:=0                      |                                             |                |
| <del>fa</del> $\neg$ Q.isEmpty()                       |                                             |                |
| u:=Q.rem()                                             |                                             | c:=c+1         |
| color[u]:=black                                        |                                             | p:= A[u]       |
| <del>fa</del> $p \neq 0$                               |                                             |                |
| v:= p->v                                               |                                             |                |
| color[v]=white                                         | color[v]=grey                               | color[v]=black |
| color[v]:=grey                                         | <i>return false</i><br><del>fa: hamis</del> | skip           |
| Q.add(v)                                               |                                             |                |
| p:=p->next                                             |                                             |                |
| return ( <del>fa</del> c=n)                            |                                             |                |